

Pink Oregon Trail

Teacher Edition — Curriculum Standards, Learning Objectives, Test Bank, and Lesson Plan

What this game is

Pink Oregon Trail is a hybrid printable-board / browser game designed for middle-school social studies. Students lead a wagon train of seven from Independence, Missouri to Oregon City between 1841 and 1852. The game models real Oregon Trail mechanics — disease, river crossings, wagon repair, food management, weather, Native trade, and the historical fork-in-the-road decisions pioneers actually faced. Every game is different, and every game teaches.

Why it exists

Sixth-grader Gabriella "Gabby" Kasdaglis designed this game with her father for her sixth-grade history class. The standard Oregon Trail games she had played felt thin: kids learned rhythms but not history. Pink Oregon Trail is built to teach the actual content of westward expansion — the geography, the human and tribal dimensions, the choices, and the costs — while still being a real game her classmates would want to play.

What's in this edition

- **Foundational philosophy** — five principles that guide every design choice (page 2)
- **Learning objectives** — measurable outcomes by content area (page 3)
- **Curriculum standards mapping** — Florida B.E.S.T. Standards (SS.5.A.6, SS.6.W.1, SS.6.G, ELA.6.R.2) with cross-cutting anchors (page 4)
- **Lesson plan** — a one-class-period flow with pre-game intro, gameplay, and debrief (page 5)
- **Test bank with answer key** — twenty-five questions and short-answer prompts pulled from the game's historical fact pool (pages 6-7)
- **Calibrated success rates** — what to expect, by difficulty and trail length (page 8)
- **Classroom features and assessment** — Class Session ID, Teacher Mode dashboard, and the "What You Learned" page printout (page 9)
- **Discussion prompts and extensions** — for class debrief and homework (page 10)

For the student-facing rules document, see [pink_oregon_trail_rules.html](#). The game itself runs from [pink_oregon_trail.html](#) in any modern browser, online or off.



What's new in v3.4 (this edition)

Since v3.3, the game added a **pre-journey outfitting screen** (kids pick a wagon, an animal team, and up to two heirlooms before naming the team — a real economic decision with reward/risk uncertainty), a **tightened economy** (starting purse \$110-\$160 with daily money depletion, fort prices roughly doubled), a **medicine-choice modal with visible survival percentages** when a member falls ill, an **always-visible "Spot N of M" indicator** in the calendar bar so kids can match the screen to the printed board, two new **36×24 print posters** (Short and Extended), and a **kid-friendly DIY map guide** for art-class extension. Also: the California ending has been removed; all teams now finish at Oregon City.

Philosophical Foundations

Five principles that guide every design choice

1. Historical fidelity over comfort

The Oregon Trail killed about one in ten emigrants in 1849. The game honors that. Players sometimes lose. Members sometimes die. We do not soften the trail, and we do not pretend everyone made it. The trail teaches its own lesson, and that lesson is part of the curriculum. About thirty thousand people are buried along the historical route, most in unmarked graves. The game treats those deaths with weight — a brief funeral scene, a journal entry, a moment of silence in the music.

2. Strategy must matter

Earlier versions of the game produced identical outcomes for skilled and careless wagons — both finished one hundred percent of the time. The current version corrects that: a careful Hunter+Doctor team reliably finishes, while a careless team without protective skills often fails. Strategy choices — which wagon and animals to outfit at Independence, which heirlooms to bring, which professions to hire, when to push on, when to stop at a fort, whether to take the Sublette Cutoff or the slower Bridger road, when to spend a precious dose of medicine and when to save it — measurably affect the outcome. Kids who pay attention finish more often than kids who don't. That's the lesson.

2a. Decisions cost something (v3.4)

Before the trail begins, every team spends from a starting purse of \$110-\$160 on a wagon, animal team, and up to two heirlooms. Every dollar spent is a dollar not available for food, repair, or medicine on the trail. Then on the trail itself, each Push On day quietly drains a small amount for tolls, oxen feed, and barter — money is never stagnant. The medicine-choice modal asks the player to weigh a single dose against a member's daily survival probability, with a clear "save the medicine" option. None of these decisions has a single right answer — but each has visible trade-offs the player can reason about. That is what economic and moral judgment under uncertainty looks like.

3. Surprise is part of the lesson

Sometimes a profession that doesn't seem useful turns out to be essential. A Musician keeping morale up across a hard week. An Artist whose sketches lift the team's spirits at Independence Rock. A Banker whose extra coin pays the Barlow Road toll the day the Columbia raft would have killed them. The narrative engine surfaces these unexpected wins so kids learn that every team member has unique gifts. Optimal team-building isn't always the right answer; diverse teams can beat "perfect" ones because of who's there when something unexpected happens. This is a real lesson about people and groups, not just about Oregon Trail tactics.

4. Knowledge pays off

The game's "Did You Know" cards aren't decoration — they're the trail's curriculum, and they're tied directly to gameplay. Students who pay attention to the cards do better on the bandit quiz at journey's end. They unlock conversational options at later events ("we remember what the fort sutler said"). They earn higher Scholar scores. Reading the cards is rewarded; ignoring them costs points. We want kids to want to learn the content because the content matters mechanically, not because we tell them it does.

5. Respect for Native peoples

Native nations were guides, traders, healers, teachers, and neighbors more often than they were antagonists. The game tells that story honestly, and includes the harder truths: the Donation Land Claim Act of 1850 explicitly excluded Native peoples, Black Americans, and Hawaiians from owning land in Oregon. The Whitman Mission was destroyed in 1847 partly because emigrant-borne measles had killed roughly half of the Cayuse children — a tragedy that came from both sides of the encounter. Pioneer journals from the period document Native generosity far more often than Native hostility.

The game treats Pioneer Americans and Native Americans as full human beings with full stories. Both populations suffered; both showed courage; both made choices that look different from this side of the Donation Land Claim Act. Students should leave the game knowing the trail was a meeting place, not a conquest.

A note on the calibration: The simulator is honest. Wagons without protective skills (no Hunter, no Doctor) finish at lower rates than the spec's averaged-everyone bands suggest, because real history was harder on unprotected wagons than averaged numbers convey. We have left this asymmetry in the game. It is a teaching moment about the difference between a population mean and the experience of any one wagon.

Learning Objectives

What students will know and be able to do

Geographic literacy

- Identify and locate the major rivers of the Oregon Trail (Missouri, Platte, North/South Platte fork, Sweetwater, Green, Bear, Snake, Columbia, Willamette) and explain how the trail followed water.
- Identify and locate the major mountain ranges (Rockies, Wasatch, Cascades, Blue Mountains, Sierra Nevada) and explain why South Pass was the critical crossing.
- Recognize the territorial map of 1848: Missouri as a state, Kansas/Nebraska/Wyoming as unorganized territory, the Oregon Country as American (post-1846 treaty) but bordering British Hudson's Bay Company territory at the 49th Parallel.
- Distinguish the Oregon Trail from the California Trail, including the Hastings Cutoff and Donner Party tragedy that grew out of choosing differently.

Historical context

- Explain why pioneers left the East — economic depression after 1837, Donation Land Claim Act prospects, religious freedom, joining family, gold (after 1848).
- Identify why mid-April was the standard departure date and how the winter clock at Day 165 created life-and-death pacing.
- Recognize that cholera, dysentery, and accidents (not violence) caused the great majority of trail deaths.
- Understand the role of Native nations as guides, traders, ferrymen, and healers — far more often than as antagonists. Identify the Pawnee, Lakota, Shoshone, Nez Perce, Cayuse, and others by their territories.
- Read the moral complexity of the Whitman Mission destruction (1847) and the Donation Land Claim Act exclusions (1850).

Source literacy

- Distinguish primary from secondary sources and recognize that pioneer women's diaries are the principal primary source for daily trail life.
- Recognize that historical sources have perspectives and biases (mountain man, fort sutler, Native trader, returning emigrant — each tells the trail differently).
- Practice weighing the trustworthiness of conflicting accounts using contextual clues.

Quantitative reasoning

- Apply distance/speed/time calculations: 12-20 miles per wagon-day, 2,000 miles total, 165-day budget. Estimate when a wagon can stop and still arrive on time.
- Manage limited resources (food, water, money, ammunition) under uncertainty.
- Reason qualitatively about risk: when is a 50-mile waterless cutoff worth seven days saved? When is it not?

Civic and moral reasoning

- Articulate why the Donation Land Claim Act's racial exclusions matter to how we evaluate the "promise" of the trail.
- Discuss the trade-offs of cutoffs and shortcuts as a metaphor for risk-and-reward decisions.
- Reflect on the meaning of leaving home for an uncertain future — a choice millions of people in history have made.

Curriculum Standards Mapping

Florida B.E.S.T. Standards (2023 revision)

Pink Oregon Trail content does not fit Florida 6th-grade Social Studies directly — that course is World History (ancient and classical civilizations of Africa, Asia, and Europe). It maps cleanly to **SS.5.A.6 (5th grade — Growth and Westward Expansion)** as a review/extension activity, and to **SS.8.A.4 (8th grade — Westward Expansion)** as preparation. Cross-cutting standards from the 6th grade curriculum apply throughout.

Primary content standards

Code	Description	Where it lives in the game
SS.5.A.6	Westward Expansion (5th grade)	Entire game; primary tagging on every Did You Know card
SS.8.A.4	Westward Expansion (8th grade — preparation/preview)	Entire game; alternate-route choices test 8th grade reasoning

Cross-cutting skill anchors (6th grade)

Code	Description	Where it lives in the game
SS.6.W.1.3	Primary/secondary sources, perspective and bias	Trail News mechanic — judging trustworthiness of sources at forts
SS.6.G.1	Geographic interpretation	Mini-map with rivers, mountains, territories; landmark recognition
SS.6.G.4	Human-environment interaction	Push-On Tax, water management, regional terrain effects
SS.6.G.6	Mapping	Expandable mini-map, physical 9-tile board, distance reasoning
ELA.6.R.2.2	Informational text comprehension	40 historical Did You Know cards plus bandit quiz
ELA.6.R.3	Integration of knowledge and ideas	"What You Learned" end-of-game retrospective; What If branch

Three demonstration levels (rubric)

For each Did You Know card or quiz item, students can demonstrate at one of three rubric levels:

- 1. Recall** — the team SAW the Did You Know card during the journey (logged in the wagon's history).
- 2. Apply** — the team made a CHOICE involving the standard's content (e.g., heeded a Native guide's advice; bypassed Whitman ruins).
- 3. Analyze** — the team got the QUIZ question RIGHT at the bandit ambush (the trail's "final exam").

The Class Session dashboard tracks all three levels per student wagon and produces a printable Standards Coverage Matrix for administration.

A note on benchmark codes

Specific sub-codes (e.g., SS.5.A.6.5) for the 17 historical facts added in v3.3.1 have not yet been verified at cpalms.org. The game tags these facts with their general standard category but leaves the exact sub-benchmark field

empty until verified. Better to ship with missing tags than fabricated ones. Teachers may wish to verify and add these codes for their administration's portfolio.

Lesson Plan

A 50-minute class period (extends to 90 minutes for Extended)

Materials needed

- Chromebooks or tablets (one per wagon team — 2-4 students per wagon recommended)
- Pink Oregon Trail loaded from local USB or nkasdaglis.github.io/pink-oregon-trail/play.html
- Student Edition rules HTML (one printed copy per team for reference)
- Optional: physical 3×2 ft printable board (9 tiles, taped together) for hybrid play
- Optional: simple markers (paper clips, beads, mini figurines) for the physical board

The 50-minute flow

1. Pre-game introduction

5-7 minutes · whole class

Project the historical map. Show students the route from Independence to Oregon City. Ask: "Why would a family in Ohio in 1848 leave everything they own and walk two thousand miles?" Field three or four answers. Reveal: free land, religious freedom, economic hardship, gold (1849+). Explain the winter clock — they have until early October before mountain snow closes the passes.

2. Game setup with Class Session ID

3-5 minutes · students setup their wagons

Have students open the game. Walk them through the setup screens together: pick Short trail (15-20 minute version), pick Medium difficulty, then **outfit at Independence** (wagon + animals + up to two heirlooms — a real economic decision; see Page 3 of the Student Edition). Then build a team of seven by choosing professions (the picker is alphabetical for quick scanning). **Turn on Teacher Mode** and have every team enter the same Class Session ID (write it on the board: e.g., "6A-2026-04-28"). This is what makes the post-game class dashboard work.

Teaching moment during outfitting: ask groups "Why didn't every team pick the most expensive wagon?" Listen for the trade-off articulation. Some will spend everything on protection and arrive broke; some will skimp on the wagon to keep cash for forts. Both can finish; both can fail.

3. Gameplay

25-30 minutes · independent team play

Teams play their wagons. Walk the room. Listen for moments worth surfacing: a team that just lost a member to cholera, a team that took the Sublette Cutoff and lost an ox. These become discussion fuel.

Watch for: students reading Did You Know cards aloud (good); skipping them to push on (cost them later); arguing about pace and rations (the strategy lesson at work).

4. End-of-journey debrief at each team

5 minutes · students review their What You Learned page

Teams print or screenshot their What You Learned page (which has the four scoring tracks, the historical facts they encountered, the choices they made, who they lost or saved, and five auto-generated discussion questions). Give them 5 minutes to read and discuss within the team.

5. Class debrief with the Class Session dashboard

8-10 minutes · whole class

Click "End Class Session" on the teacher screen. Project the Class Dashboard. Walk through:

- *Aggregate Outcomes*: how many wagons finished out of how many; how many members started vs. arrived. Compare the class to the historical 1-in-10 death rate.
- *Class Heatmap*: which Did You Know facts every wagon saw vs. which ones nobody saw — gaps for tomorrow's teaching.
- *Aggregate Decisions*: did most teams take the Sublette Cutoff or play it safe through Fort Bridger? Why?
- *Top Discussion Prompts*: pick two or three to discuss aloud.

The Class Session dashboard prints to a single letter page — file it in the unit's evidence portfolio.

For Extended journey (90-minute block)

Use the same flow, but allow 50-60 minutes for gameplay. Extended adds 25 more historical waypoints and brings the journey to roughly 40-50 minutes of play. Recommended for repeat sessions or block schedules.

Test Bank — Part 1

Multiple-choice questions with answer key. Pulled from the game's historical fact pool.

1. How long was the Oregon Trail?

- A. About 500 miles
- B. About 2,000 miles
- C. About 5,000 miles
- D. About 200 miles

Answer: B — about 2,000 miles. (Comparable to walking from New York City to Denver.)

Standard: SS.6.G.1, SS.5.A.6

2. Why did wagon trains have to leave Independence by mid-April?

- A. To avoid summer heat in the mountains
- B. To beat the rainy season in Oregon
- C. To meet the U.S. Army for protection
- D. To reach Oregon before winter snows closed the mountain passes

Answer: D. Prairie grass needed time to grow tall enough to feed the oxen, and emigrants needed to clear the Rockies before October. The Donner Party left too late in 1846.

Standard: SS.6.G.4 (human-environment interaction)

3. What was the #1 killer on the Oregon Trail?

- A. Cholera
- B. Native American attacks
- C. Wolf attacks
- D. Wagon accidents

Answer: A — cholera. Spread through dirty water at busy river crossings. A healthy person could feel a stomach ache at breakfast and be dead by sundown.

Standard: SS.5.A.6, ELA.6.R.2.2

4. About what fraction of Oregon Trail emigrants died on the journey?

- A. About 1 in 100
- B. About 1 in 10
- C. About 1 in 3
- D. Almost all of them

Answer: B — about 1 in 10, or 30,000 people total over 1841-1869. Most are buried in unmarked graves.

Standard: SS.6.W.1.3 (interpreting historical data)

5. About what fraction of Oregon Trail emigrants were children?

- A. Less than 1 in 20
- B. About 1 in 3
- C. About half

D. Almost none — the trail was too dangerous

Answer: B — about one-third. Children walked alongside the wagons, picked up buffalo chips for fires, and gathered wild fruit.

Standard: SS.5.A.6, ELA.6.R.2.2

6. What did pioneers use for fuel when there were no trees on the prairie?

- A. Coal they bought at trail forts
- B. Tumbleweeds rolled into bundles
- C. Dried buffalo droppings ("buffalo chips")
- D. Leftover wood scraps from the wagon

Answer: C. Buffalo chips burned hot, almost smokeless, and almost odorless once dried. Children gathered them by the bushelful.

Standard: SS.6.G.4

7. Why did most pioneers use oxen instead of horses to pull wagons?

- A. Horses were illegal to take across the Mississippi River
- B. The U.S. government required oxen for all wagon trains
- C. Horses couldn't drink the water west of Independence
- D. Oxen were stronger, calmer, cheaper, ate prairie grass, and were less likely to be stolen

Answer: D. Oxen made about 2 mph; a wagon train made 12-20 miles on a good day.

Standard: SS.6.G.4 (human-environment interaction)

8. What landmark did pioneers carve their names into?

- A. Independence Rock
- B. Chimney Rock
- C. Mount Hood
- D. Soda Springs

Answer: A. The 130-foot granite dome in Wyoming. More than 5,000 names from the 1840s and 1850s are still visible in the stone today. Reaching it by July 4th meant you were on schedule for Oregon.

Standard: SS.6.G.1, SS.6.G.6

9. What was the Continental Divide and why did pioneers cross it at South Pass?

- A. The line where free states ended; pioneers had to register with the U.S. Army
- B. The U.S.-Canada border; pioneers had to show passports
- C. The line where rivers flow east toward the Atlantic vs. west toward the Pacific. South Pass was a wide, gentle saddle that wagons could cross easily.
- D. The boundary between Oregon and California; pioneers had to choose

Answer: C. South Pass is so gentle that pioneers often topped it without realizing they had crossed the spine of the continent.

Standard: SS.6.G.1, SS.6.G.6

10. Who was Sacagawea, and how did she help the Lewis & Clark expedition that came before the wagon trains?

- A. A Cheyenne queen who ruled the Plains
- B. A Lemhi Shoshone woman who translated and identified edible plants for the expedition
- C. A Spanish explorer's daughter

D. An American mapmaker

Answer: B. She traveled with Lewis and Clark from 1805-1806, about forty years before the wagon trains. Her presence with a baby signaled to other tribes that the expedition came in peace.

Standard: SS.5.A.6, SS.6.W.1.3

11. What was the Donation Land Claim Act of 1850?

- A. A law banning all settlement in Oregon
- B. A treaty between the U.S. and Native nations
- C. A law requiring emigrants to pay \$50 to settle in Oregon
- D. A law that gave free land in Oregon to settlers who farmed it for four years — but explicitly excluded Native peoples, Black Americans, and Hawaiians

Answer: D. A married white couple received 640 acres; a single white settler received 320 acres. The Act's racial exclusions are part of the trail's story.

Standard: SS.5.A.6, SS.8.A.4 (civic/legal context)

Test Bank — Part 2

Continued from Page 6

12. Why did the Whitman Mission end in tragedy in 1847?

- A. A flood washed the mission buildings away
- B. Measles brought by emigrants killed many Cayuse children, leading to violence
- C. The Whitmans abandoned it for California gold
- D. The U.S. Army closed it down

Answer: B. About half of the Cayuse children died of measles brought by emigrant travelers. On November 29, 1847, a Cayuse party killed Marcus Whitman, Narcissa, and twelve others. After 1848, emigrants knew to bypass the ruins.

Standard: SS.5.A.6, SS.6.W.1.3 (multiple perspectives)

13. About how much did it cost a family of four to outfit for the Oregon Trail in 1848?

- A. About \$20 — almost anyone could afford it
- B. About \$700 to \$1,000 — more than two years' wages
- C. About \$50,000 — only the wealthy could go
- D. Free — the U.S. government paid for everything

Answer: B. Most pioneers were not the very poor; they were middling farmers selling everything they owned for a chance at land in Oregon.

Standard: SS.5.A.6, SS.8.A.4

14. What was the Sublette Cutoff?

- A. A federal law against cutting trees on the trail
- B. An alternate route opened in 1844 that saved 70 miles by skipping Fort Bridger — but crossed 50 miles of waterless desert
- C. The shortcut to California gold
- D. The official name for the Whitman Mission detour

Answer: B. By 1849, about half of emigrants chose it. Those who guessed wrong on water sometimes lost oxen and family members to thirst.

Standard: SS.6.G.4 (human-environment interaction)

15. What historical sources do historians most use to learn about daily life on the Oregon Trail?

- A. Government surveys of every wagon train
- B. Newspaper articles published in Oregon City
- C. The diaries kept by pioneer women
- D. Photographs taken at every stop

Answer: C. Pioneer women on the trail did most of the cooking, child care, and medical work — and many wrote diaries. Their journals tell us most of what we know.

Standard: SS.6.W.1.3 (primary source identification)

Short-answer / extended-response prompts

16. About 1 in 10 Oregon Trail emigrants died on the journey. Pick one of the major causes of death (cholera, river drowning, accidental gunshot, exposure, starvation) and explain in 2-3 sentences why it killed so many.

Sample answer: Cholera spread through contaminated water at busy river crossings where many wagon trains shared springs. The disease moved fast — a healthy person could feel sick at breakfast and die by sundown — and emigrants didn't know to boil water for two minutes (the only sure prevention they had).

Standard: SS.5.A.6, ELA.6.R.2.2

17. Pick a profession from the game (Hunter, Doctor, Carpenter, Cook, Scout, Native Guide, Trapper, Farmer, Priest, Banker, Merchant, Musician, Artist, Cowboy) and explain in 2-3 sentences how that profession contributed to a wagon train's chances of survival.

Sample answer (Cook): A Cook stretched the team's food rations by about 22% per meal, meaning the wagon could go further on the same flour. Cooks also recognized signs of food spoilage, saleratus poisoning, and bad water — saving lives that wagons without a Cook lost.

Standard: SS.6.G.4

18. The Donation Land Claim Act of 1850 explicitly excluded Native peoples, Black Americans, and Hawaiians from owning land in Oregon. In 4-5 sentences, explain how this exclusion changes the way we should think about the "promise" of the Oregon Trail. Who benefited and who didn't?

Sample answer: The trail offered free land — but only to people the U.S. government considered eligible. White settlers received 320 to 640 acres each; the Cayuse, Nez Perce, Walla Walla, and other peoples who had lived on that land for centuries were displaced. Black Americans and Hawaiians who walked the same trail couldn't claim what white settlers received. The "promise" was real for some people and a closed door for others, and the wealth that grew from those land claims reflects that uneven start.

Standard: SS.5.A.6, SS.8.A.4, civic reasoning

19. Pioneer women wrote most of the diaries that survive about Oregon Trail daily life. Why do you think their accounts, more than men's, became the principal historical record?

Sample answer (any thoughtful response): Women had different daily roles (cooking, nursing, child care) that observed the camp from a different angle than the men's hunting and driving. Many were literate and trained to keep family records. They wrote during long evening pauses while men were tending oxen or repairing wagons. Their diaries also tracked births, deaths, and marriages — the facts of family life that mattered to them and that historians now use to understand what daily trail life felt like.

Standard: SS.6.W.1.3, ELA.6.R.3

20. The game lets you choose to take the Sublette Cutoff (save 7 days, but 50 miles waterless) or go through Fort Bridger (slower but reliable resupply). In a real wagon master's journal entry of 4-5 sentences, explain which one your wagon chose and why, in the voice of someone making that decision in 1849.

Sample answer (any plausible voice): "We took the Bridger road. Five days out from Independence Rock our barrels were already low, and we had a sick child. The trapper at Big Sandy said the cutoff would save us seven days, but he had a flask in his hand and his eyes were rheumy. I trust the road I can see. Bridger has water and a blacksmith. We will lose the days but keep the family."

Standard: SS.6.W.1.3 (perspective), SS.6.G.4 (decision under environmental constraint)

Calibrated Success Rates — Detail

What to expect by team composition, difficulty, and trail length

These numbers come from running each game configuration five hundred times in a Monte Carlo simulator and averaging the results. Two reference team types are used to bracket the range — your students' actual teams will fall between, depending on team composition and play strategy.

v3.4 calibration note: The numbers below were calibrated against the v3.3 economy. v3.4 raised starting money to \$110-\$160, doubled fort prices, added daily money depletion, and introduced loadout costs. The doubled prices roughly cancel the doubled purse, so finish rates should remain near the bands shown — but a class that consistently finishes well above or below these numbers might warrant a fresh `#calibrate3=500` run before the next session.

Strong team (HD) = wagon with both a **Hunter** (food supply via Hunt action and passive income) AND a **Doctor** (87% cholera save rate, sickness mitigation). Represents a well-prepared wagon.

Weak team (MA) = wagon with a **Musician** AND **Artist** — both morale-only specialists, no protection. Represents a wagon that focused on culture and not survival skills.

Short Journey — 25 stops, 15-20 minute play

Difficulty	Strong team finish %	Weak team finish %	Strong team avg members lost	Notes
Easy	97-98%	83-87%	0.5-0.8 of 7	Mild years, 1841-43; almost everyone makes it
Medium	85-88%	36-43%	0.9-1.1 of 7	1849 standard; the historical average
Hard	62-68%	10-14%	1.1-1.4 of 7	1852 cholera year; about 4 in 10 fail

Extended Journey — 50 stops, 35-50 minute play

Difficulty	Strong team finish %	Weak team finish %	Strong team avg members lost	Notes
Easy	96-98%	58-66%	2.6-2.8 of 7	Long journey accumulates damage
Medium	72-77%	14-20%	3.0-3.2 of 7	1849; about 1 in 4 strong teams fails
Hard	38-43%	1-5%	2.9-3.2 of 7	1852 harsh year; about 6 in 10 wagons fail

Strategy gap — how much do choices matter?

The "strategy gap" is the difference in finish rate between the strong and weak reference teams. v3.2 had a 0% gap (both finished 100% of the time). v3.3 puts the gap at 11-58 percentage points depending on cell, with a mean of about 42 percentage points. The bigger the gap, the more strategy matters; the more strategy matters, the more the game teaches.

How to read the math

If you run one Medium / Short class with 8 wagons and one or two don't finish, that is consistent with the calibration (about 80% of wagons make it on Medium / Short). If three or four wagons don't finish, your wagons may have been mostly weak teams or they played carelessly — both are teachable moments.

The historical 1849 death rate was about one person in ten. If a class of 8 wagons ($8 \times 7 = 56$ emigrants) loses about 5-6 members across all wagons by Day 165, that's the historical average. The Class Session dashboard surfaces this number directly.

Classroom Features and Assessment

Teacher Mode, Class Session ID, and the printable evidence dashboard

Teacher Mode

Toggle Teacher Mode on the Difficulty setup screen. This unlocks:

- The Class Session ID input (one ID per class period)
- The "Print Teacher Report" button at end of game (per-wagon detailed sheet)
- The Standards Demonstrated section in each wagon's What You Learned page
- Aggregate access to the Class Dashboard at end of class period

The Class Session ID

Pick a Session ID at the start of class — for example, "6A-2026-04-28" or "PER3-NICKK-2024". Have every student wagon enter the same ID. The game stores all wagons' results in localStorage keyed by that ID. Once every wagon has finished its journey, click "View Class Dashboard" on any wagon's end-screen.

The Class Dashboard — five sections

1. **Standards Coverage Matrix** — which Florida benchmarks were exercised across the class. Print and file with the unit's evidence portfolio.
2. **Class Heatmap** — Did You Know facts split into "every wagon saw" / "some wagons saw" / "no wagon saw" columns. The "no wagon saw" column is your tomorrow-morning lesson plan.
3. **Aggregate Decisions** — count of pivotal choices made (which fork did most teams pick at the Sublette Cutoff?). Discussion fuel.
4. **Aggregate Outcomes** — total members started, alive, lost, average days to Oregon, causes of loss breakdown. Compare to the historical 1-in-10.
5. **Top Discussion Prompts** — eight to ten prompts auto-generated from this specific class session's data.

The "What You Learned" page (per wagon)

Each wagon's individual end-of-game printable. Six sections:

1. *Your Trail in Numbers* — the four scoring tracks (Steward, Provider, Scholar, Pathfinder) as side-by-side bars with interpretation.
2. *What History You Met* — gallery of every Did You Know card seen.
3. *The Choices You Made* — retrospective of every fork-in-the-road decision.
4. *The People You Lost (or Saved)* — per-member status with epitaph.
5. *Discussion Questions for Your Teacher* — five prompts auto-generated for this specific journey.
6. *Standards Demonstrated* — Florida benchmarks the journey covered with one-line "I Can" statements.

Trail Journal preserved as a collapsible accordion at the bottom (Gabby's mom asked for it).

What If — The Roads Not Taken

For every alternate-route choice the wagon made (Sublette / Bridger, Barlow / Columbia raft, Whitman bypass / visit ruins, etc.), the end-of-game page also shows what the OTHER choice would have given. Useful for "what would have happened if..." discussion.

Assessment options

- **Formative** — observe gameplay; listen for which DYK cards students reference; watch how they reason about the Sublette risk warnings
- **Per-wagon summative** — score the wagon's bandit quiz performance (5 questions, multiple-choice); the score is built in
- **Whole-class summative** — Class Dashboard's standards coverage matrix becomes the artifact
- **Extended** — assign one of the short-answer prompts from the test bank as homework following the game

Discussion Prompts and Extensions

For class debrief, homework, and follow-up units

Class debrief prompts

1. About one in ten Oregon Trail emigrants died on the journey. Our class lost ____ of ____ members across all wagons. How does our class's number compare to the historical rate? What does that tell us?
2. Which Did You Know facts did NOBODY's wagon see this game? What might that say about the random paths the trail offered, and what does it tell us about doing this game multiple times?
3. Most teams in our class chose [Sublette / Bridger]. Why do you think most teams went that way? What did the teams who chose differently know that the others didn't?
4. If you ran the same wagon train again with the same team — same difficulty, same length — would the result be the same? Why or why not?
5. Pioneer women wrote most of the surviving diaries. If our class kept a journal of THIS specific game, what would future students learn about US, three centuries from now?

Homework / extension prompts

- **Primary source reading:** Have students find one real pioneer diary entry online (Library of Congress, Oregon Historical Society, or Oregon-California Trails Association archives) and compare it to what their wagon experienced in the game. Where do they agree? Where do they disagree?
- **Geography map:** On a printed blank Oregon Trail map, have students mark the rivers, mountain passes, and forts in their game's order, then write a 5-sentence summary of why each location mattered.
- **Native nation profile:** Pick one of the Native nations encountered in the game (Pawnee, Lakota, Shoshone, Nez Perce, Cayuse, Walla Walla, etc.). Research what happened to that nation between 1840 and 1880. What changed? What did the wagon trains contribute to that change?
- **Donation Land Claim Act:** Read the 1850 Donation Land Claim Act language about race and write a one-page response on what that exclusion means for how we evaluate the "promise" of the Oregon Trail. What do students with Mexican-American, Asian-American, Black, or Native heritage learn about their families' history from that document?
- **Family history:** Have each student ask one family member: "Has anyone in our family ever made a long-distance move because of opportunity or hardship?" Compare to pioneer experience.

Suggested unit extensions

- **Day 2 follow-up:** The "no wagon saw" facts from the Class Heatmap become the next day's reading list. Print the cards and assign one per student to research.
- **Cross-curricular with Math:** Use the trail data — 12-20 miles per day, 2,000 miles total, 165-day budget — for word problems on rate, distance, and time.
- **Cross-curricular with English:** Have students write a 5-paragraph wagon-master journal in period voice (1840s English: no "vibes," no modern slang). Use the test bank prompt #20 as a template.
- **Cross-curricular with Geography/Civics:** Map the Donation Land Claim boundaries and discuss the connection between land law and demographic change.

- **Replay challenge:** Have students play Hard / Extended and report back. About 2 in 10 wagons will finish — that means most students will lose. Discuss: how does it feel to "lose" a history game? What does that teach you about the people who actually walked the trail?

Resources and attribution

Pink Oregon Trail uses public-domain content. Major sources include:

- National Park Service — Whitman Mission NHS, Fort Laramie NHS, Lewis & Clark NHT, South Pass NHL
- Bureau of Land Management — National Historic Trails Office
- Oregon-California Trails Association — historical records, trail maps
- Smithsonian National Museum of American History — material culture (wagon design, period objects)
- Oregon Historical Society — Donation Land Claim Act records
- National Women's History Museum — pioneer women's diaries
- Library of Congress — public-domain photographs (~50 historical images embedded in the game)

The game is open and free. Students may keep their wagon journal printouts as historical-thinking artifacts.

Companion documents

- **[pink_oregon_trail_rules.html](#)** — student-facing rules booklet (9 pages, prints to letter paper). Updated for v3.4.1.
- **[docs/board_v33_poster_short_36x24.html](#)** — print-ready 36×24 poster with the 25 Short-trail stops. Send to a print service for an Engineer Print, mount on foam board.
- **[docs/board_v33_poster_extended_36x24.html](#)** — same, with all 50 Extended-trail stops.
- **[docs/make_your_own_map.html](#)** — kid-friendly art-class extension. Thirteen-step DIY guide for making a 22×28 poster-board map with watercolor and crayons. Letter paper print.



Pink Oregon Trail v3.4.1 — Teacher Edition. Designed by Gabriella Kasdaglis with her father, for her sixth-grade history class — and beyond.